

Timbre Branch Architecture V2

Interface preserving accuracy and research quality revision

Music Genre Classification Research Project
MTG Jamendo 7324 track subsample
Version 2 | September 2026

Decision V2 preserves the complete V1 public tensor contract while revising training, target preprocessing, loss balancing, validation, and research evaluation. The default V2 model retains the single 35-output head.

The purpose of V2 is to improve descriptor fidelity and genre usefulness without requiring changes to the shared encoder interface, other branches, fusion token dimensions, or genre head.

Version 2 invariant	Value
Branch input	h_audio FloatTensor [B,128]
Branch output	z_timbre FloatTensor [B,35]
Default neural head	V1 single output head retained
Fusion projection	Fusion owned Linear(35,64)
Primary target change	Training-fitted log transform for spectral_flatness_mean
Primary loss change	Group-balanced masked Smooth L1

1 Version transition

V2 is an interface-preserving revision. It does not change how the timbre branch connects to the shared encoder or fusion. It does change the meaning of one standardized target coordinate, the optimization schedule, and the way descriptor families contribute to the auxiliary loss.

Design element	V1	V2	Compatibility
Input	[B,128]	[B,128]	Unchanged
Output	[B,35]	[B,35]	Unchanged
Default output head	Single Linear(64,35)	Single Linear(64,35)	Unchanged
Fusion	Linear(35,64)	Linear(35,64)	Unchanged
Epoch budget	Up to 50	Up to 50 with scheduler	Training only
Learning rates	Single optimizer rate	Lower shared encoder rate	Training only
Loss	Cell-averaged Smooth L1	Group-balanced Smooth L1	Shape safe
Flatness target	Raw then z-score	Log then z-score	New preprocessing state
Ablations	Required in principle	Required run protocol	Evaluation only

1.1 Public contract

Input:	h_audio	FloatTensor [B,128]
Output:	z_timbre	FloatTensor [B,35]
Supervision target:	d_true_standardized	FloatTensor [B,35]
Supervision mask:	valid_mask	BoolTensor [B,35]
Fusion projection:	owned by fusion	Linear(35,64)

Any implementation that changes these shapes, reorders the 35 descriptors, returns a branch-owned fusion token, or exposes h_audio directly to fusion is not Architecture V2.

2 Baseline evidence from results.json

V2 is motivated by the completed V1 joint run rather than by an assumed weakness. The run used 5127 training tracks, 1099 validation tracks, and 1098 held-out test tracks. The best checkpoint occurred at epoch 20, which was also the final recorded epoch.

Baseline measurement	Observed result	V2 interpretation
Best epoch	20 of 20	Training ended while the selection metric was still improving
Validation genre macro AP	0.766	Strong baseline and primary scheduler signal
Test genre macro AP	0.736	Reasonable held-out generalization
Test genre macro F1	0.622 at threshold 0.5	Useful baseline; threshold tuning is separate
Test genre micro F1	0.636 at threshold 0.5	Reference for full-model comparisons
Test timbre standardized MAE	0.318	Best continuous branch error in this run
Test timbre standardized RMSE	0.561	Some concepts retain substantial residual error
Test timbre macro R squared	0.701	Most declared timbre variance is recoverable
Timbre observed cells	38430	All 1098 test tracks x 35 concepts were evaluated

2.1 Evidence from the training trajectory

Epoch	Validation macro AP	Validation timbre loss	Observation
1	0.358	0.2335	Initial representation
5	0.635	0.1693	Large early improvement
10	0.726	0.1376	Continued learning
15	0.754	0.1174	Smaller improvements
20	0.766	0.1146	Best AP and lowest recorded timbre loss

Because both genre AP and timbre loss improved through the last epoch, V2 retains a 50-epoch ceiling with early stopping rather than assuming epoch 20 is sufficient.

3 Timbre fidelity diagnosis

Aggregate timbre performance is strong, but per-feature results are uneven. V2 targets the weak flatness coordinate while preserving well-performing concepts and the complete 35-value contract.

Descriptor	Test R squared	Standardized MAE	Finding
spectral_contrast_mean	0.936	0.138	Strongest spectral concept
mfcc_02_mean	0.931	0.149	Very strong envelope estimate
mfcc_01_mean	0.922	0.166	Very strong envelope estimate
spectral_rolloff_mean	0.881	0.226	Strong high-frequency extent estimate
spectral_flatness_mean	0.090	0.466	Primary V2 preprocessing target
mfcc_13_mean	0.475	0.453	Weak higher-order MFCC mean
mfcc_05_mean	0.486	0.441	Weak mid-order MFCC mean
spectral_centroid_std	0.495	0.475	Weak temporal-variation estimate

3.1 Genre context

Genre	Test average precision	Test F1
classical	0.834	0.748
electronic	0.688	0.574
folk	0.693	0.607
hiphop	0.867	0.779
jazz	0.608	0.361
rock	0.724	0.662

The full-model scores do not prove how much timbre caused the genre result. Jazz is the weakest genre by AP and F1, which reinforces the need for timbre-only and no-timbre comparisons rather than attributing any genre result to timbre from the joint run alone.

4 Default Version 2 neural architecture

```
h_audio [B,128]
-> Linear(128,128)
-> LayerNorm(128)
-> GELU
-> Dropout(0.20)
-> Linear(128,64)
-> GELU
-> Linear(64,35)
-> z_timbre [B,35]
```

The V2 primary experiment deliberately retains the V1 single output head. This isolates the effect of better preprocessing, loss balancing, scheduling, and optimization. Architecture capacity is not changed in the primary comparison.

Component	Output	Version 2 role
128 to 128 projection	[B,128]	Adapt the pooled shared representation
LayerNorm and GELU	[B,128]	Stable nonlinear mapping
Dropout 0.20	[B,128]	Regularization; initially unchanged
128 to 64 projection	[B,64]	Internal timbre state
Single 64 to 35 head	[B,35]	Ordered standardized concepts

4.1 Descriptor order remains fixed

Indices	Descriptor family	Count
0 to 6	Spectral shape	7
7 to 8	Harmonic and noise	2
9 to 21	MFCC means	13
22 to 34	MFCC standard deviations	13

The output names and order remain those declared in `timbre_branch` constants. Only the preprocessing of `spectral_flatness_mean` changes in V2.

5 Version 2 target preprocessing

5.1 Training-only flatness transform

V1 test fidelity for `spectral_flatness_mean` was substantially weaker than for the other descriptors. V2 addresses its positive, skewed scale before ordinary standardization.

```
epsilon_flatness = fitted or declared positive constant
x_flatness_log = log(max(x_flatness_raw, epsilon_flatness))
z_flatness = (x_flatness_log - train_mean_log) / train_scale_log
```

The transform configuration is fitted or fixed without using validation or test values. Inverse explanations first undo z-scoring and then apply `exp` to return to the original flatness scale.

Requirement	Version 2 rule
Fit population	Training split only
Affected concept	<code>spectral_flatness_mean</code> only
Other 34 concepts	Ordinary training-fitted z-score
Stored checkpoint state	Transform name, epsilon, mean, scale, and feature order
Explanation output	Inverse transformed original acoustic units
Validation and test	Use frozen training transformation unchanged

5.2 Preprocessing version identity

```
preprocessing_version = "timbre_v2_log_flatness_zscore_v1"
```

A V1 checkpoint must not be loaded as V2 without an explicit migration. The tensors have equal shape, but the flatness coordinate uses a different numerical representation.

6 Group-balanced masked Smooth L1

```
L_shape = masked_smooth_l1(pred[:,0:7], target[:,0:7])
L_harmonic= masked_smooth_l1(pred[:,7:9], target[:,7:9])
L_mfcc = masked_smooth_l1(pred[:,9:35], target[:,9:35])
L_timbre = (L_shape + L_harmonic + L_mfcc) / 3
```

Equal group weighting prevents the 26 MFCC coordinates from dominating the seven spectral-shape and two harmonic-noise coordinates merely because the group contains more columns. Each group continues to use validity masks and returns a differentiable zero when it has no observed cells.

7 Version 2 optimization protocol

Setting	Version 2 primary protocol
Maximum epochs	50
Early stopping	Validation metric with patience 8
Optimizer	AdamW
Shared encoder learning rate	Lower than branch and fusion learning rates
Timbre branch learning rate	Higher task-head rate
Scheduler	ReduceLROnPlateau
Scheduler factor	0.5 initial recommendation
Scheduler patience	3 initial recommendation
Minimum learning rate	1e-6 initial recommendation

7.1 Differential learning rates

```
shared_encoder: 1e-4
timbre_branch: 5e-4
fusion_head: 5e-4
```

These are V2 starting values rather than universal optima. The required relationship is that the shared encoder receives a lower learning rate than newly trained task-specific components. All selected values must be recorded in the run configuration.

7.2 Scheduler and model selection

```
scheduler = ReduceLROnPlateau(mode="max", factor=0.5,
                               patience=3, min_lr=1e-6)
scheduler.step(validation_macro_average_precision)
```

The best validation checkpoint is restored before the held-out test set is evaluated once. The run log records every learning-rate change, stopping epoch, selected checkpoint epoch, and loss term.

8 Training sequence

1. Validate the 128-dimensional shared-encoder input and 35-dimensional target order.
2. Fit the V2 preprocessing state using training rows only.
3. Train the unchanged single-head timbre bottleneck within the full model for up to 50 epochs.
4. Apply ReduceLROnPlateau and early stopping using validation data only.
5. Restore the best validation checkpoint.
6. Evaluate the test partition once and save per-concept metrics.

9 Amendment rationale

Each V2 amendment responds to either the observed V1 results or the need to isolate scientific causes. The sequence below is part of the experiment design, not an unordered list of options.

Amendment	Why it is included
1 Keep h_audio [B,128]	Preserves compatibility with the shared encoder and avoids changing every branch integration point.
2 Keep z_timbre [B,35]	Preserves the named concept bottleneck, fusion projection, feature order, and explanation interface.
3 Train up to 50 epochs with early stopping	The best checkpoint occurred at the final epoch 20 while validation AP and timbre loss were still improving.
4 Add ReduceLROnPlateau	Later gains were smaller and validation metrics fluctuated; reducing the rate can refine the model without extending high-rate updates.
5 Lower the shared encoder learning rate	The encoder serves four branches. Smaller updates reduce the risk that timbre optimization damages representations needed elsewhere.
6 Use group-balanced Smooth L1	The current 26 MFCC coordinates outnumber the seven spectral-shape and two harmonic-noise values, so a cell average favors MFCCs by count.
7 Log-transform spectral flatness	Flatness had R squared 0.090 and RMSE 0.895, far below most concepts. A recorded positive-value transform tests whether target scale is the obstacle.
8 Retain the single output head first	Keeping model capacity fixed allows performance changes to be attributed to preprocessing and training rather than a simultaneous head redesign.
9 Run timbre-only and no-timbre ablations	Joint genre accuracy cannot establish the timbre branch contribution. Matched ablations measure its isolated and incremental value.
10 Test grouped heads separately as V2-G	A separate run measures descriptor-family specialization without confounding the primary V2 comparison and requires a new checkpoint.
11 Run shape and gradient tests first	Automated tests prevent an internal improvement from silently breaking [B,128] input, [B,35] output, fusion, masks, or gradient flow.

10 Required ablation protocol

V2 requires controlled comparisons to determine whether improved descriptor fidelity creates useful genre information rather than only a lower auxiliary loss.

Run	Active timbre route	Research question
V2 Full	Predicted 35 concepts	Primary complete-model result
V2 Timbre only	Predicted timbre concepts only	How much genre signal is present in timbre?
V2 No timbre	Timbre fusion mask disabled	What is timbre's incremental contribution?
V2 Oracle timbre	Extracted target concepts	What is the upper bound with perfect concepts?
V2 Shuffled timbre	Concepts assigned to other tracks	Do correct track-level concepts matter?

10.1 Primary comparisons

Incremental timbre value = $AP(V2 \text{ Full}) - AP(V2 \text{ No timbre})$

Prediction bottleneck gap = $AP(V2 \text{ Oracle timbre}) - AP(V2 \text{ Full})$

Report overall macro and micro average precision and F1, plus per-genre changes. Timbre concept metrics include standardized MAE, RMSE, R squared, and per-descriptor results.

11 Shape and gradient contract tests

- Input rank other than two is rejected.
- Input width other than 128 is rejected.
- A valid [B,128] input produces finite [B,35] output.
- Feature names contain exactly 35 unique entries in the published order.
- Fusion projects [B,35] to [B,64] and applies its mask after projection.
- The adapter rejects fusion_token and h_audio bypass fields.
- Gradients propagate from timbre loss through every V2 timbre parameter.
- Gradients propagate from genre loss through the 35-dimensional concept route.
- Checkpoint reload reproduces predictions and preprocessing inversion.

12 Separate grouped-head experiment

Grouped internal heads are not part of the primary V2 comparison. They form a separate experiment, identified as V2-G, after the single-head V2 result is established.

```
shared trunk [B,64]
-> spectral_shape_head      -> 7
-> harmonic_noise_head      -> 2
-> mfcc_mean_head           -> 13
-> mfcc_std_head            -> 13
-> concatenate in fixed order -> z_timbre [B,35]
```

Property	V2-G decision
Public input	[B,128], unchanged
Public output	[B,35], unchanged
Fusion	Linear(35,64), unchanged
Internal parameters	Changed; requires a new checkpoint
Comparison	Matched split, seed, preprocessing, loss, and optimizer budget
Adoption rule	Promote only if validation and repeated-seed evidence improve

12.1 Why it is separated

Changing preprocessing, loss, scheduling, and head structure simultaneously would prevent attribution. The primary V2 run changes training behavior while keeping the V1 head. V2-G then measures the additional effect of descriptor-family specialization.

13 Explainability requirements

- Always expose standardized predictions and inverse-transformed original units.
- Mark weak-fidelity concepts as uncertain rather than treating every concept equally.
- Run descriptor and group interventions through the same fusion pathway used in training.
- Do not claim causal properties of music from model-internal interventions.
- Retain the strict prohibition on direct `h_audio` access by fusion.

14 Version 2 reproducibility package

Artifact	Required content
Architecture report	This V2 specification and version identifier
Source revision	Git commit and tag for all branch and fusion code
Target schema	Exact 35 names and ordering
Preprocessor	Log-flatness settings plus all training means and scales
Data split	TRACK_ID to train, validation, or test assignment
Run configuration	Learning rates, scheduler, epochs, patience, seed, and loss weights
Checkpoint	Model, optimizer, preprocessing, and architecture metadata
Results	Per-epoch logs, per-concept metrics, genre metrics, and ablations

14.1 Version identity rules

Change	Version treatment
Epoch, scheduler, or learning-rate tuning with the V2 preprocessor	V2 training revision
Grouped internal heads with the same public contract	V2-G experiment and new checkpoint
Changing the target transformation again	New preprocessing revision
Adding, deleting, or reordering concepts	New major architecture version
Changing the input to temporal encoder tensors	New major architecture version
Adding a direct audio shortcut to fusion	Invalid strict concept-bottleneck model

15 Version 2 acceptance statement

Timbre Branch Architecture V2 is an interface-preserving development of V1. The primary model continues to accept `h_audio` [B,128] and return `z_timbre` [B,35] through the original single output head. V2 adds training-only log preprocessing for spectral flatness, group-balanced masked Smooth L1 loss, differential learning rates, ReduceLROnPlateau, early stopping, mandatory ablations, and shape and gradient verification.

V2-G remains a separately reported internal-head experiment. Neither V2 nor V2-G may change the public input, output, feature order, or fusion ownership without declaration of a new major version.